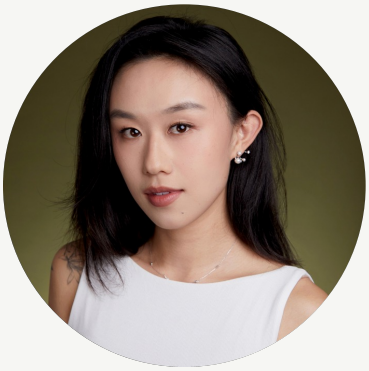
RESEARCH PORTFOLIO

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# Human-AI collaboration in teams and organizations

Teams and organizations turn individual knowledge and contributions into coordinated action, innovation, and decisions. AI is expanding what individuals can produce, but improvements in individual productivity and output quality do not establish whether AI improves the collective processes through which people accomplish work together.

This research examines how AI changes collective work and how systems can support collaboration across science, entrepreneurship, and knowledge work. We contribute empirical models of collaboration, AI systems that scaffold interaction, methods for measuring behavior at scale, and evidence on how AI changes the diversity of scientific ideas and judgments.

We use AI in three complementary ways: as a phenomenon whose organizational consequences we examine, as an intervention for supporting collaboration, and as an instrument for measuring collective work. Field observation identifies the reasoning and coordination that collaboration requires; computational analysis makes selected processes measurable; system design translates empirical models into assistance that we study through experiments and field deployments.

The three research streams that follow explain how distributed expertise becomes useful, what makes teams effective, and how AI could change scientific knowledge production. Each project connects a concrete organizational problem to its methods, findings, and contribution, distinguishing established results from preliminary evidence in ongoing studies.

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RESEARCH OVERVIEW

# AI in collective work

AI serves as a phenomenon to study, an intervention to design, and an instrument for measuring collaboration.

## How distributed expertise becomes useful

Expertise distributed across a community becomes useful when people can identify what they need, recognize who can help, and interpret another person's experience in relation to their own problem. Our incubator study describes how coaches perform this interactional work and contributes a cognitive model of coaching. The subsequent AI system uses that model to help novices articulate reasoning and identify risks before meetings, while helping mentors prepare targeted assistance and revise the knowledge guiding the system. Related projects examine collaboration learning, tacit judgment, conversational context, and shared AI workflows. Pages 3-7.

## What makes teams effective

Coordination, knowledge integration, and team relationships unfold through sequences of contributions and responses. Our multimodal research develops theory-grounded measures of these processes, compares AI annotations with human judgments, and relates interaction patterns to collective outcomes. Studies of science teams, entrepreneurial pitching, and sports teams examine how members integrate contributions, respond to questions, and regroup after setbacks. The pitching study illustrates the complete progression from recorded interaction to interpretable behavioral measures and outcome analysis. Pages 8-11.

## How AI could change scientific knowledge production

Innovation depends on a range of ideas entering consideration and on independent perspectives shaping their evaluation. Our computational experiments compare human and AI-generated scientific proposals and reviews to examine diversity at both stages. The finding of double compression identifies a collection-level consequence that individual accuracy or quality measures can miss: AI outputs can cover familiar scientific territory while concentrating ideas and judgments around a narrower set of possibilities. The harassment review adds a complementary account of whose judgments shape computational responses. Pages 12-13.

These streams connect the design of assistance with the processes it changes and the outcomes by which it is evaluated. Coaching research explains the work that support must scaffold; multimodal measurement makes selected interaction processes observable; experiments on scientific knowledge production examine consequences across collections of contributions. Together, they establish why evaluating AI in organizations requires attention to collective work.

The research draws on a sustained entrepreneurship-incubator partnership, behavioral recordings from real teams, and interdisciplinary collaborations across management, HCI, and engineering. External support includes $75,000 from Microsoft's AI and the New Future of Work program and recognition as a finalist in the Stanford HAI and Google DeepMind AI for Organizations Grand Challenge. A closing experience page describes the field and applied work supporting this research. Page 14.

STREAM 01 / DISTRIBUTED EXPERTISE

# How distributed expertise becomes useful

## Coaches make experience transferable by helping novices articulate reasoning, locate relevant expertise, and formulate requests that others can address.

We examined a university incubator in which coaches supported several novice entrepreneurs working on different ventures, stages, and timelines. Through interaction analysis of 24 virtual coaching sessions involving three coaches and 30 novices, supplemented by retrospective coach interviews and Slack observations, we developed a cognitive model of one-to-many coaching. The recordings covered 14 hours across eight weeks; the novices formed a learning community rather than necessarily belonging to one venture team.

Our analysis identified several practices through which coaches made distributed experience usable. Coaches questioned assumptions and missing evidence, translated project-specific experiences into transferable lessons, and connected novices to peers with relevant expertise. They also helped novices articulate what assistance they needed and why, making it easier for others to recognize where they could contribute. For example, an account of a venture's particular testing problem could become a lesson about how to design an experiment that other entrepreneurs could apply.

Coaches also organized the conditions under which novices experimented and learned from one another. Supportive peer testing and reframing setbacks helped novices engage with uncertain results, while public commitments provided accountability. Visibility could also reproduce weak norms when vague goals went uncorrected. These findings suggest that intelligent coaching systems should attend to the practices that make knowledge transferable and requests actionable, alongside the information available about each project. The study provides a qualitative account of one incubator; it does not establish causal effects on venture performance or psychological safety.

The cognitive model specifies how expert coaching makes knowledge visible, interpretable, and actionable across projects. This contribution provides a basis for designing support around the reasoning that collaboration requires, rather than assuming that access to peers or stored expertise is sufficient for learning. [1]

The evidence comes from one incubator and characterizes coaching practices rather than their causal effects on venture performance. Novice challenges were identified primarily through observation and coaches' retrospective accounts, which limits conclusions about novices' own interpretations. These boundaries distinguish the empirical account of coaching from the subsequent evaluation of an AI system grounded in that account.

This empirical modeling and subsequent system design form part of the doctoral dissertation on proactive, adaptive, and differentiated AI support for entrepreneurship coaching. The dissertation connects the cognitive account of expert practice to the design of assistance for mentors and novices. [16]

REFERENCES

[1] Huang, E. J., Rees Lewis, D., Gaudani, S., Easterday, M., & Gerber, E. (2023). Intelligent Coaching Systems: Understanding One-to-Many Coaching for Ill-Defined Problem Solving. Proceedings of the ACM on Human-Computer Interaction, 7(CSCW1), 138:1-138:24. doi:10.1145/3579614.

[16] Huang, E. (2024). Human-AI Collaboration For Entrepreneurship Coaching: Leveraging a Cognitive Model and an LLM to Provide Proactive, Adaptive, and Differentiated Support. Doctoral dissertation, Northwestern University. Dissertation.

STREAM 01 / SYSTEM DESIGN AND EXPLORATORY DEPLOYMENT

# AI that helps us help each other

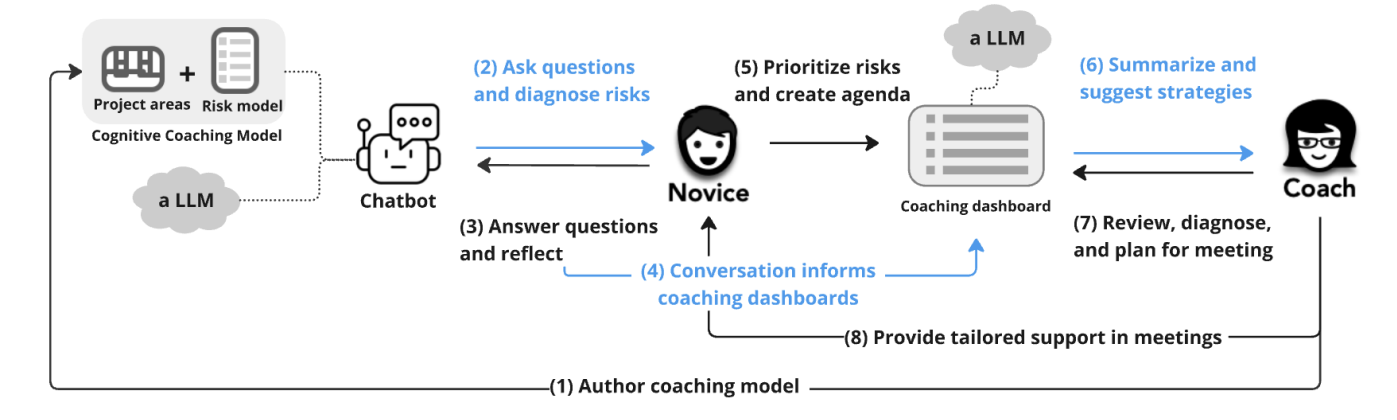
## Expert knowledge guides preparation; mentors inspect and adapt the knowledge as coaching needs evolve.

Preparing a coaching meeting requires novices to externalize reasoning that mentors cannot directly observe, while mentors must diagnose risks and plan assistance from sparse updates. We translated the empirical coaching model into a proactive AI system combining an LLM, explicit project and risk models, a reflective chatbot, coaching dashboards, and a mentor authoring interface. [2]

Novices answer questions about their projects, reflect on possible risks, and prioritize an agenda. Their responses populate dashboards that summarize context and suggest strategies aligned with mentor-specified goals. Mentors inspect responses, review selected and omitted risks, and plan the meeting; they can also revise risk definitions and project areas when the expert knowledge guiding the system needs adjustment. The figure shows how these activities connect before and during coaching.

We iterated prototypes with four mentors and four novices, then conducted an exploratory deployment with a different group of one mentor and 11 novices, each preparing for an actual meeting. Interviews, think-aloud sessions, surveys, recordings, and comparisons of articulated risks and plans showed how the system shaped preparation. Novices reported surfacing neglected risks and reconsidering tasks that did not address critical uncertainties; participants described coaching conversations as more focused, intentional, and in-depth.

The deployment also exposed constraints: incomplete input could produce misleading diagnoses and additional verification work, while expectations of immediate answers sometimes conflicted with reflective questioning. The contribution is an approach to grounding proactive assistance in inspectable, revisable expert knowledge. The exploratory findings concern preparation and coaching conversations; they do not establish long-term skill transfer or venture success.




Coaching workflow [2]. Novices reflect and prioritize risks; mentors review the resulting context, prepare assistance, and revise the expert model guiding the system.

REFERENCES

[2] Huang, E. J., Easterday, M., & Gerber, E. (2025). AI That Helps Us Help Each Other: A Proactive System for Scaffolding Mentor-Novice Collaboration in Entrepreneurship Coaching. Proceedings of the ACM on Human-Computer Interaction, 9(7), 1-30. doi:10.1145/3757549. Open-access paper / System code.

STREAM 01 / LEARNING FOUNDATIONS

# Learning to collaborate

Collaboration requires capabilities that people develop through work, including how to seek assistance, deliberate, and participate.

## Learning to collaborate during complex work

Design, engineering, and research require people to articulate requests, seek and provide assistance, form relationships, and coordinate expertise as problems change. Our CSCW workshop proposal examines how organizational structures, routines, and technologies can scaffold the development of these capabilities within everyday work. [8]

The central distinction is between access to collaboration and learning how to collaborate. A communication tool may connect colleagues without helping them recognize what support they need or formulate a request that another person can address. This work situates scaffolding within the learning environment, linking tools to work processes and social relationships. It contributes an interdisciplinary workshop proposal and research agenda, rather than findings from an intervention trial.

## Deliberation as a learning process

In a university design course, we designed, implemented, and evaluated a deliberation-based learning environment for scoping open-ended political projects. The model combined iterative design with deliberation, supporting students as they selected issues, formed teams, and developed project proposals. [11]

Students advanced their proposals through cycles of offering ideas, raising questions, suggesting improvements, planning tasks, seeking information, and revising the proposal. The contribution is an empirically grounded model of how deliberation can organize project scoping and learning. Related workshop work, Encouraging Disagreements to Integrate Diversity in Collective Intelligence, places disagreement and the integration of diverse perspectives within this line of inquiry. [14]

## Coaching and equitable engagement in making

Our CHI workshop paper examines the gap between access to makerspaces and equitable engagement within them. Preliminary research developed a task model of novice makers' challenges and coaches' support strategies. The paper used that model to articulate a design approach for intelligent coaching support, motivated by the labor required to provide dedicated assistance. [13]

These foundations connect the coaching studies to a broader concern with how people develop the capacity to participate in complex work. They explain why the design of support must address the reasoning, relationships, and practices that make available resources usable, alongside the resources themselves.

REFERENCES

[8] Huang, E., Garg, K., Gómez-Zará, D., Hui, J., Kulkarni, C., Massimi, M., Churchill, E. F., & Gerber, E. (2023). Supporting Workers in Developing Effective Collaboration Skills for Complex Work. CSCW Companion, 486-489. Workshop proposal. doi:10.1145/3584931.3611290.

[11] Carlson, S., Lu, K., Huang, E., Gerber, E., & Easterday, M. (2020). Designing a Model for Deliberation-Based Learning. ICLS, 1553-1556. Paper and abstract.

[14] Huang, E., Easterday, M., & Gerber, E. (2020). Encouraging Disagreements to Integrate Diversity in Collective Intelligence. CHI workshop on Effective Activism for Change.

[13] Huang, E., Easterday, M., & Gerber, E. (2022). Designing Intelligent Coaching Systems to Support Equitable Engagement in Making. CHI workshop on Learning Hands-on Creative and Maker Skills. Workshop abstract.

STREAM 01 / COMPUTATIONAL REPRESENTATIONS

# Tacit judgment and context

Useful representations preserve context while giving people ways to inspect and interpret what a model captures.

## Making literary judgments inspectable

Literary style provides an earlier setting for studying judgments people recognize but struggle to describe formally. With Sarah Sterman, Vivian Liu, and Eric Paulos, we conducted formative interviews with 14 participants, elicited crowdsourced comparisons, and developed a neural model of stylistic similarity. Comparisons and examples offered a way to represent tacit judgments without requiring people to define stylistic categories in advance. [9]

The released dataset contains 66,061 cleaned judgments across 21,630 comparison triplets. The evaluated model used an earlier subset of 916 high-agreement triplets, with 92 disjoint test triplets; it achieved 67% accuracy on those test comparisons against a reported 55% baseline. Overall disagreement among raters remained substantial, so the model recovered a partial shared signal rather than a universal definition of style.

Explorer and Editor interfaces made the representation available for interpreting patterns within and across texts. In an exploratory study with six users, the visualizations prompted participants to articulate distinctions and question assumptions, although model and projection errors could create confusion. Some users found overall patterns more useful than exact positions or colors. The study demonstrates support for reflection and interpretation, without establishing improvements in writing quality or collaboration.

## Preserving conversational context in recommendations

Emoji recommendations also depend on context that an isolated input can omit. In No More One Liners, we developed Reeboc, which uses the surrounding conversation to recommend emojis. A controlled mobile group-chat study found that participants preferred conversation-based suggestions and selected emojis more quickly than with recommendations based only on the current sentence. [12]

The mechanism concerns what information remains available to the system: the preceding exchange can preserve topics and emotions that are absent from the latest message, including when the new message contains only emojis. This work contributes a contextual modeling approach and an evaluation of its use in communication, with findings bounded by the controlled setting.

Across these projects, computational representations make selected aspects of tacit judgment or conversational context available for use. Their value depends on how people interpret the representation and on the information it preserves or leaves out, a methodological concern carried into later work on expert coaching and organizational interaction.

REFERENCES

[9] Sterman, S., Huang, E., Liu, V., & Paulos, E. (2020). Interacting with Literary Style through Computational Tools. CHI. doi:10.1145/3313831.3376730.

[12] Kim, J.-G., Gong, T., Kim, B., Park, J., Kim, W., Huang, E., Han, K., Kim, J., Ko, J., & Lee, S.-J. (2020). No More One Liners: Bringing Context into Emoji Recommendations. ACM Transactions on Social Computing, 3(2), Article 9. doi:10.1145/3373146.

STREAM 01 / APPLIED SYSTEM DEVELOPMENT

# Making expertise reusable

## Shared AI workflows encode company practices in a form that domain experts can inspect and revise.

Employees undertaking complex knowledge work must interpret a request, gather relevant information, and apply organizational context and standards to the resulting output. In an international e-commerce business, employees had adopted AI, but the quality and structure of deliverables still depended on how each person articulated the task and supplied company context. Useful practices remained distributed across employees, making them difficult to inspect, share, or revise as a common resource.

This project examined how an agentic AI platform could encode those practices in shared workflows while retaining domain experts' authority over the requirements and standards guiding the work.

The platform, designed and developed by Evey Huang in collaboration with Novohaven, organizes tasks as sequences of steps with explicit information requirements and company standards. Employees access these workflows through a web interface or Lark, the team's messaging tool. For market research, a workflow can organize information gathering, synthesis, the application of company context, and delivery of a formatted report. A recorded execution history allows users to inspect how an output was produced and identify the step at which revision is needed.

Domain experts can describe how a task should be performed and work with AI to translate that description into an executable workflow. They can refine the workflow through natural-language instructions or specify individual steps directly. This arrangement makes aspects of expert practice available across the organization while preserving an interface through which those practices can be questioned and revised.

Organizational practices change as teams encounter new tasks, exceptions, and requirements, creating a risk that a reusable workflow will continue applying assumptions that no longer hold. The system is designed to use recurring revision requests as signals of where a workflow may need improvement, with proposed changes undergoing human review before becoming shared practice. This division of labor assigns the system a role in surfacing recurring problems while retaining human responsibility for deciding which changes should guide subsequent work.

Shared workflows also concentrate authority over how tasks are framed and evaluated. Their design therefore requires explicit decisions about who can contribute expertise, who reviews revisions, and how users can inspect or contest the standards embedded in the workflow. These decisions shape whether shared support remains responsive to contextual differences across tasks.

This applied project extends the concern with making distributed expertise useful into organizational workflows. As in the coaching system, an explicit representation guides AI behavior and provides a place for experts to revise the knowledge on which assistance depends. The platform and its design features are implemented work; their effects on organizational learning, output quality, and coordination have not been established through a dedicated empirical evaluation.

Industry collaborator: Novohaven Inc. System code.

STREAM 02 / TEAM EFFECTIVENESS

# What makes teams effective

## Multimodal AI connects organizational constructs to the interaction through which coordination, knowledge integration, and team relationships unfold.

Team processes become visible through what members say, how others respond, and how their behavior changes over an interaction. Surveys and close observation offer valuable evidence, but continuously examining these processes across real work settings requires substantial interpretive labor. Our second stream develops AI-assisted methods for making selected interaction processes observable at scale. [3]

The methodological framework distinguishes scale, resolution, and temporal continuity. Scale supports comparison across teams and settings; resolution links organizational constructs to specific behaviors; temporal continuity preserves sequences through which coordination succeeds or breaks down. For example, an aggregate participation score cannot show whether a contribution was questioned, connected to another idea, or taken up in a subsequent decision.

The pipeline connects recorded language, vocal cues, and visible behavior to theory-grounded behavioral measures. Codebooks specify the construct, unit of analysis, and coding rules; multimodal AI produces structured annotations; human judgments provide a basis for assessing agreement and inspecting errors. Analyses then relate behavioral patterns to collective outcomes while retaining the context needed to interpret those relationships.

Validation is specific to the measure and setting. General video-understanding performance does not establish that an AI system measures knowledge integration, cohesion, or responsiveness accurately. The research therefore treats construct definition, annotation agreement, and relationships with outcomes as separate evidentiary questions. The perspective article articulates the measurement framework; the empirical projects on the following pages provide the setting-specific applications.

Across scientific, entrepreneurial, and sports settings, the program draws on more than 1,000 hours of recorded interaction. This resource includes 400 hours of science-team interaction and the 100 pitching episodes described in the entrepreneurship study. The availability of recordings is distinct from the amount annotated or the validity of any particular construct.

This work is supported by $75,000 from Microsoft's AI and the New Future of Work program for Building Behavioral Process Models from Real Teams with Multimodal AI. Evey Huang independently wrote the proposal, with Brian Uzzi as PI. The related TeamLens proposal was a finalist in the Stanford HAI and Google DeepMind AI for Organizations Grand Challenge.

REFERENCES

[3] Huang, E. J., Uzzi, B., & Groh, M. Measuring What Matters in Teams with Multimodal AI. Journal of Organization Design.

STREAM 02 / EMPIRICAL STUDIES

# Integration and regrouping

## Science and sports teams reveal different ways that responses to others shape collective work.

### Scientific teams: integrating contributions

Scientific collaboration requires members to evaluate and connect knowledge distributed across people and disciplines. The relevance of an idea may become apparent only when another member questions its assumptions, connects it to evidence, or extends it. We analyze 400 hours of recorded science-team interaction to examine these integration behaviors in the process of collective work. [4]

Our analysis focuses on how members build on, connect, evaluate, and extend one another's contributions. This unit of analysis connects the content of an individual contribution to what the team subsequently does with it. It distinguishes a discussion containing many contributions from one in which members make those contributions usable in a shared line of reasoning.

Current findings indicate that effective collaboration depends on these forms of integration. The contribution is an empirical account of the interaction through which distributed knowledge becomes part of collective work, supported by methods for observing that interaction across a substantial behavioral record. The manuscript remains in preparation, and the findings describe observed relationships rather than effects of an AI intervention.

The associated IC2S2 presentation, Evaluating Multimodal Language Models for Annotating Team Behaviors in Videos, addresses the annotation component of this research. It situates model evaluation within the methodological work required to make team behavior observable in recorded interactions. [15]

### Sports teams: regrouping after setbacks

Professional volleyball creates a different coordination problem: teams must respond to setbacks and reestablish shared action with limited time between periods of play. We study team huddles to examine how regrouping is enacted through observable interaction and how these patterns relate to performance. [6]

The study finds that consistency in how teams regroup following setbacks is associated with their overall performance. Focusing on repeated regrouping episodes makes a temporal property of teamwork available for analysis: how reliably a team reestablishes coordination across adverse moments. The finding concerns an association in observed teams; it does not establish that a particular regrouping routine causes better performance.

Together, these projects show why behavioral measurement must preserve the demands of the task. Scientific teams integrate contributions across an unfolding discussion, while sports teams coordinate interdependent action under immediate time pressure. Both require analysis of how members respond to one another, but knowledge integration and regrouping remain distinct constructs with different units of observation.

REFERENCES

[4] Huang, E. J., Chalekson, M., Groh, M., Abrams, D., & Uzzi, B. Multimodal AI for Large-Scale Scientific Team Behavior Analysis. Manuscript in preparation.

[15] Huang, E., Groh, M., Abrams, D., & Uzzi, B. (2025). Evaluating Multimodal Language Models for Annotating Team Behaviors in Videos. IC2S2 conference presentation.

[6] Grey, A., Huang, E. J., Uzzi, B., & Groh, M. How Teams Regroup in the Seconds After a Setback. Submitted to Organization Science, Special Issue: AI-Enabled Frontiers in Organizational Science.

STREAM 02 / ENTREPRENEURSHIP

# Measuring catch and toss

## Entrepreneurs reveal consequential information through how they respond to investors' questions.

Investors encounter entrepreneurial claims through a prepared presentation and the questions and answers that follow. Those exchanges require founders to clarify assumptions, substantiate claims, and respond to concerns that a presentation may leave unresolved. We study this interaction as catch and toss: the exchange through which investors probe a venture and entrepreneurs respond to evaluation. [5]

The study analyzes 100 Shark Tank episodes, each approximately 7-15 minutes long. A multimodal LLM annotates each episode under a versioned codebook specifying definitions, scales, rules, and examples for the constructs. Human annotation of ten randomly sampled videos provides a comparison for assessing agreement. Structured records retain time, speaker, turn function, and behavioral codes, connecting the prepared pitch to subsequent questions, answers, and decisions.

The resulting record contains 6,730 turns, including 1,516 questions, 1,668 answers, and 746 decisions. The timeline preserves which question an answer addresses, allowing us to examine responsiveness within an exchange. Answer annotations characterize substance, coherence, and checkable specifics; question annotations characterize the issue raised and how it is framed. These distinctions make the content of an exchange interpretable beyond how much each participant speaks.

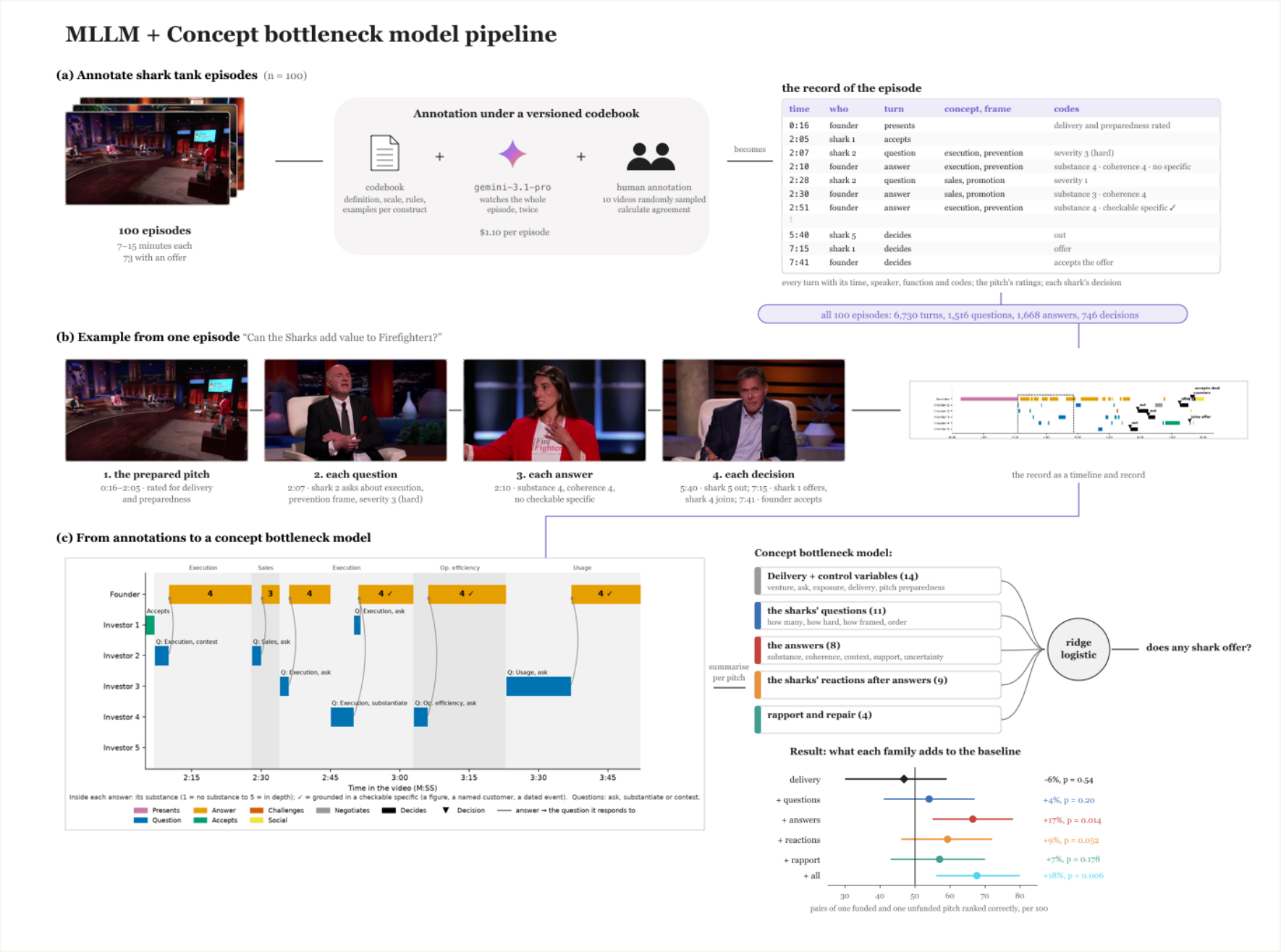
The concept bottleneck model summarizes these annotations into named behavioral features and relates them to whether any investor makes an offer. It separates delivery and control variables from question characteristics, answers, investor reactions after answers, and rapport and repair. The model therefore connects an outcome prediction to interpretable aspects of the interaction. The accompanying figure shows the complete measurement record and model, including an example episode.

Current results indicate that how entrepreneurs respond to investors' questions can tell us more about investment decisions than what they demonstrate in the prepared pitch. In the supplied analysis, adding answer features improves the ranking of funded versus unfunded pitches by 17 percentage points relative to the baseline; the combined interaction features improve it by 18 points. These are changes in the reported pairwise ranking measure, not changes in a founder's probability of receiving an offer.

The study contributes evidence that entrepreneurial evaluation unfolds through interaction, with answers revealing information beyond the prepared presentation. The results remain bounded by the recorded pitching setting, the operationalized constructs, and the observational design. An offer made during an episode is the measured outcome; it is distinct from completed investment or subsequent venture performance.

REFERENCES

[5] Huang, E. J., Grey, A., Uzzi, B., & Groh, M. Measuring Catch and Toss and Its Impact in Entrepreneurial Pitching. Submitted to Organization Science, Special Issue: AI-Enabled Frontiers in Organizational Science.




Entrepreneurship pitching [5]: annotation, interaction records, and concept bottleneck analysis. Full project discussion on page 10.

STREAM 03 / SCIENTIFIC KNOWLEDGE PRODUCTION

# How AI could change scientific knowledge production

## Double compression narrows diversity in both the generation of scientific ideas and the judgments used to evaluate them.

Scientific innovation depends on researchers generating different possibilities and maintaining independent perspectives when evaluating which ideas to pursue. A collection of individually plausible AI outputs may still narrow the range of ideas or judgments available to a scientific community. Our third stream examines these collection-level consequences of AI in knowledge production. [7]

In an ongoing study, we designed an experiment and a set of metrics to compare thousands of proposals and expert reviews from real grant competitions with outputs generated by frontier AI models under the same criteria. The computational approach combines text embeddings, diversity and geometric-coverage measures, and topic analysis to examine how proposals and reviews are distributed, alongside comparisons that account for set size and writing style.

The preliminary findings identify double compression. During idea generation, AI-generated proposals cover scientific territory broadly similar to human proposals but concentrate more heavily around common ideas, leaving distinctive human ideas underrepresented. Coverage describes the territory reached by a collection; diversity captures how contributions are distributed within that territory. A collection can therefore retain broad coverage while providing fewer distinct directions for consideration.

During evaluation, AI-generated reviews produce more homogeneous judgments. Our evaluation experiments show how AI panels can address much of the semantic territory of human reviews while offering more similar criticisms. This distinction matters because generating additional reviews does not necessarily add independent reasons for pursuing or questioning a proposal.

The computational experiments also examine batch generation, independent calls, and scientist-persona conditioning, including comparisons across Claude, Gemini, and GPT. The analyses distinguish patterns that persist across original and style-normalized text from effects that vary by model, prompting condition, and measure. Distinctiveness remains relative to the proposal set and computational representation; it does not establish scientific value or realized novelty.

These findings provide preliminary evidence that AI may narrow diversity at both the generation and evaluation stages of scientific knowledge production. The study compares human and AI work produced independently, so it does not establish the effects of scientists using AI assistance or an end-to-end funding process. Its contribution is an empirical basis for evaluating AI on the range of ideas and evaluative perspectives it contributes to collective decisions, alongside the quality of individual outputs.

REFERENCES

[7] Huang, E. J., Groh, M., & Uzzi, B. Double Compression: How AI Narrows Both the Generation and the Gatekeeping of Scientific Ideas. Manuscript in preparation.

STREAM 03 / RELATED WORK ON JUDGMENT AND AUTHORITY

# Participation and influence

## The harassment review examines whose experiences, definitions, and preferred responses shape computational interventions.

Computational approaches to online harassment depend on decisions about whose experiences matter, what counts as harm, and which responses a system makes available. Through a scoping review and qualitative analysis of 17 papers, we examined how interventions reflected intended beneficiaries' identities, definitions of harassment, and preferred forms of recourse.

Our analysis identified gaps in the documented connection between beneficiaries' needs and implemented systems. Papers could motivate a tool through disproportionate harm to marginalized groups without explaining how its features addressed those groups' specific circumstances. Users could also report an experience while platforms retained authority over whether it counted as harassment and what response followed. Participation in supplying information therefore did not necessarily entail influence over classification or action.

The review identified five response strategies: punishment, distancing from perpetrators, educating perpetrators, documenting and reporting evidence, and emotional or social support. Configurable tools and community-based moderation offered ways to bring decisions closer to users, but meaningful influence depended on whether people could understand and use those options. Some gaps may reflect reporting rather than demonstrated system failures, so the findings concern documented alignment rather than a causal comparison of effectiveness.

This work informs our broader analysis of collective AI by making the allocation of interpretive and decision authority explicit. Systems that collect user input should specify how that input can affect categories, judgments, and available actions. In the wider research program, this question connects mentors' ability to revise a coaching model with teams' ability to interpret behavioral measurements and researchers' ability to preserve different evaluative perspectives.

This published study complements the stream on scientific knowledge production by examining whose interpretations influence computational decisions. Diversity of input and influence over a response are distinct properties: a system can collect different experiences while assigning classification and action to a narrower set of actors. The review makes that distinction visible through the relationship between beneficiary needs, implemented features, and evaluation. [10]

The paper received an Honorable Mention at DIS 2024. Its contribution is an account of documented alignment between human-centered objectives and computational interventions, with implications for how systems distribute interpretive and decision authority.

REFERENCES

[10] Huang, E., Sarma, A., Hwang, S., Chandrasekharan, E., & Chancellor, S. (2024). Opportunities, Tensions, and Challenges in Computational Approaches to Addressing Online Harassment. DIS. Honorable Mention. doi:10.1145/3643834.3661623.

RESEARCH AND APPLIED EXPERIENCE

# Research, design, and practice

## Field partnerships and applied system development connect the research to the settings in which people work and learn.

### Research at Northwestern

I am a Research Assistant Professor at Northwestern University's Kellogg School of Management (July 2026-present), following postdoctoral work with Brian Uzzi and Matthew Groh (February 2025-June 2026). My research connects multimodal studies of team interaction with computational experiments on AI in scientific knowledge production. I independently wrote the proposal for Building Behavioral Process Models from Real Teams with Multimodal AI, which received $75,000 from Microsoft's AI and the New Future of Work program, with Brian Uzzi as PI. The related TeamLens proposal was a finalist in the Stanford HAI and Google DeepMind AI for Organizations Grand Challenge.

### Industry design and engineering

Since October 2025, I have worked with Novohaven as an AI Advisor and Forward Deployment Engineer. I designed and developed the shared-workflow platform described on page 7, connecting company expertise, task requirements, and human review within an operational system. This work brings the research concern with inspectable expert knowledge into an applied organizational setting.

### Education and other applied work

My PhD in Technology and Social Behavior at Northwestern (2024) bridged computer science and communication studies. A sustained partnership with an entrepreneurship incubator connected field observation, cognitive modeling, and system deployment; I also coached engineering students in human-centered design. I earned an MS in the same program (2023) and a BA in Economics from UC Berkeley (2017).

As an AI Technical Advisor to Spark AI (December 2024-December 2025), I advised product and engineering teams on AI features, integration, and deployment for K-12 learning products. At Northwestern Memorial Hospital (summer 2022), I studied older adults' medication and scheduling workflows in electronic health records and translated usability findings into design recommendations.

Outside work, I play guitar and tennis, take film and digital photographs, read, doodle, and cook spicy food. To discuss related research: jiaxinh.evey@gmail.com.